Tensor model of the psyche

based on social evidence and resource constraints

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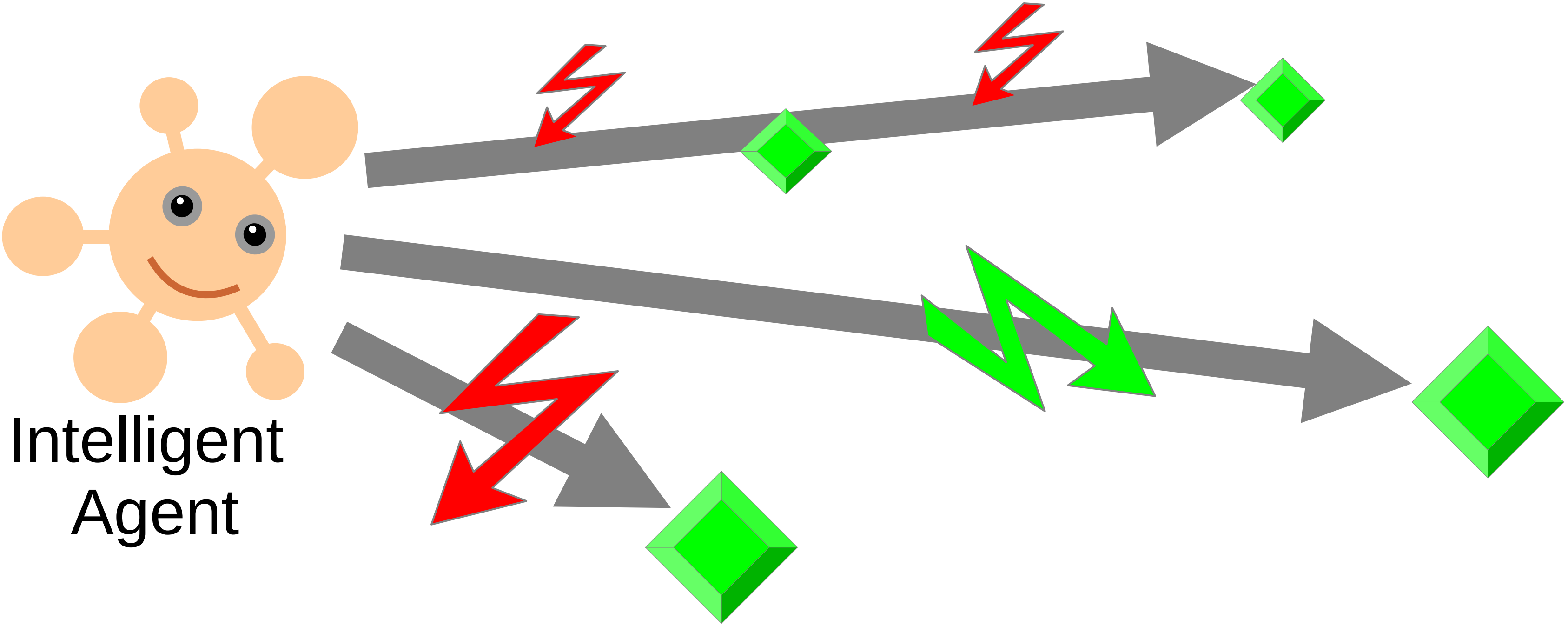
https://agirussia.org

General Intelligence:

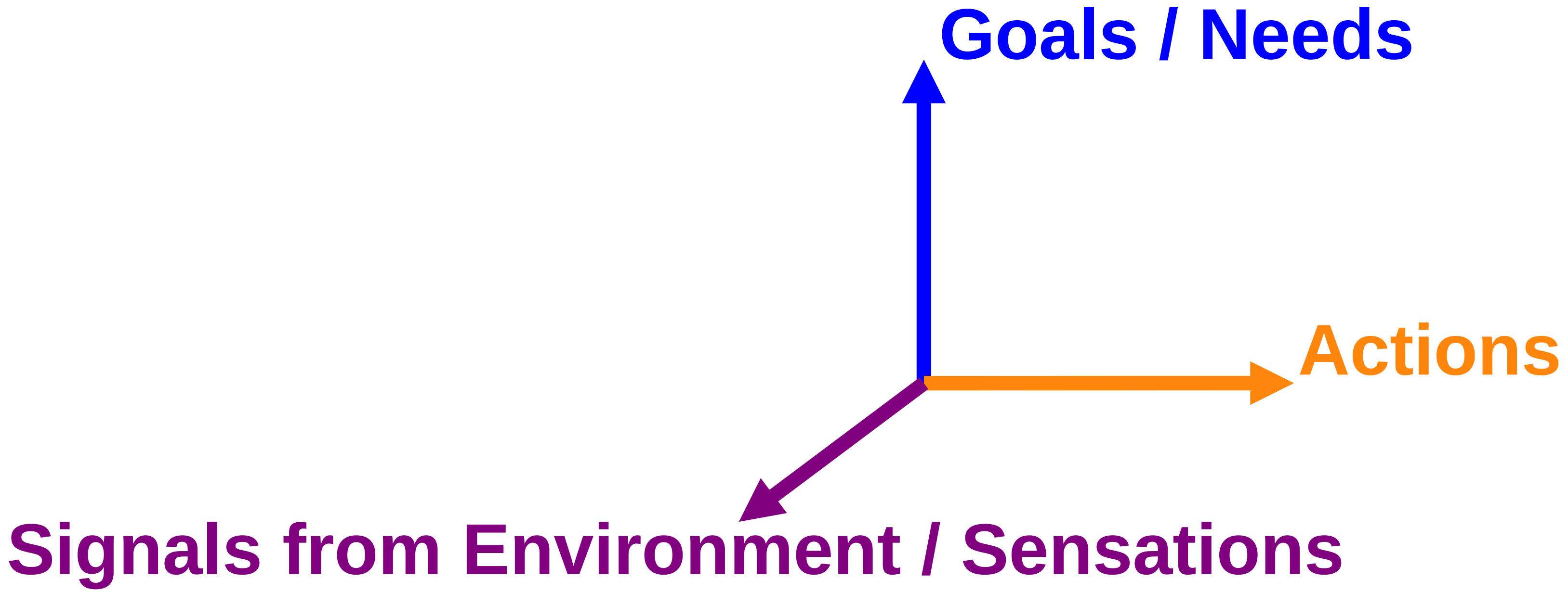
Reaching complex goals in different complex environments, using limited resources

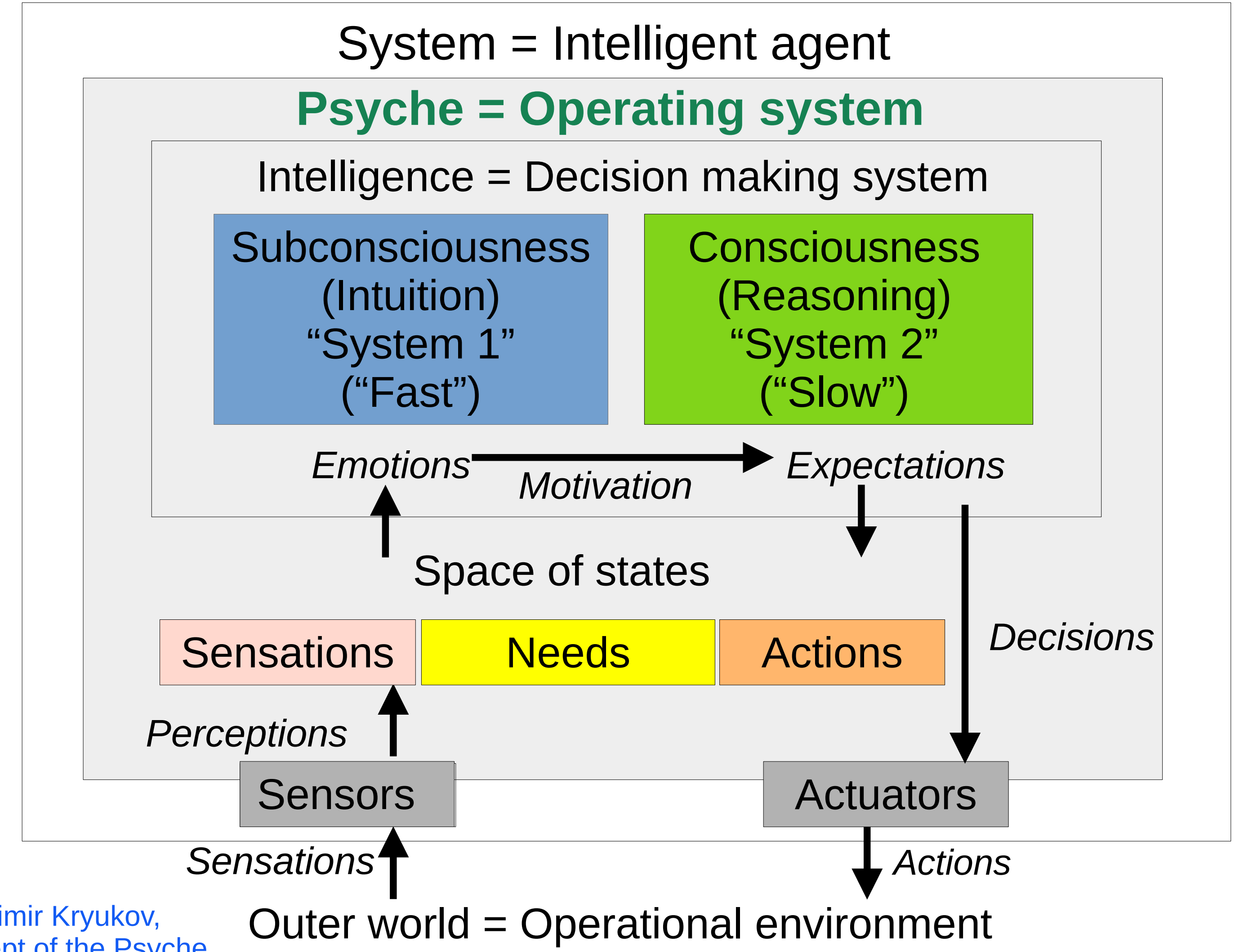
Ben Goertzel + Pei Wang + Shane Legg + Marcus Hutter



General Intelligence:

Reaching complex goals in different complex environments, using limited resources by means of complex actions





Anton Kolonin & Vladimir Kryukov,
Computational Concept of the Psyche,
https://arxiv.org/abs/2603.15586

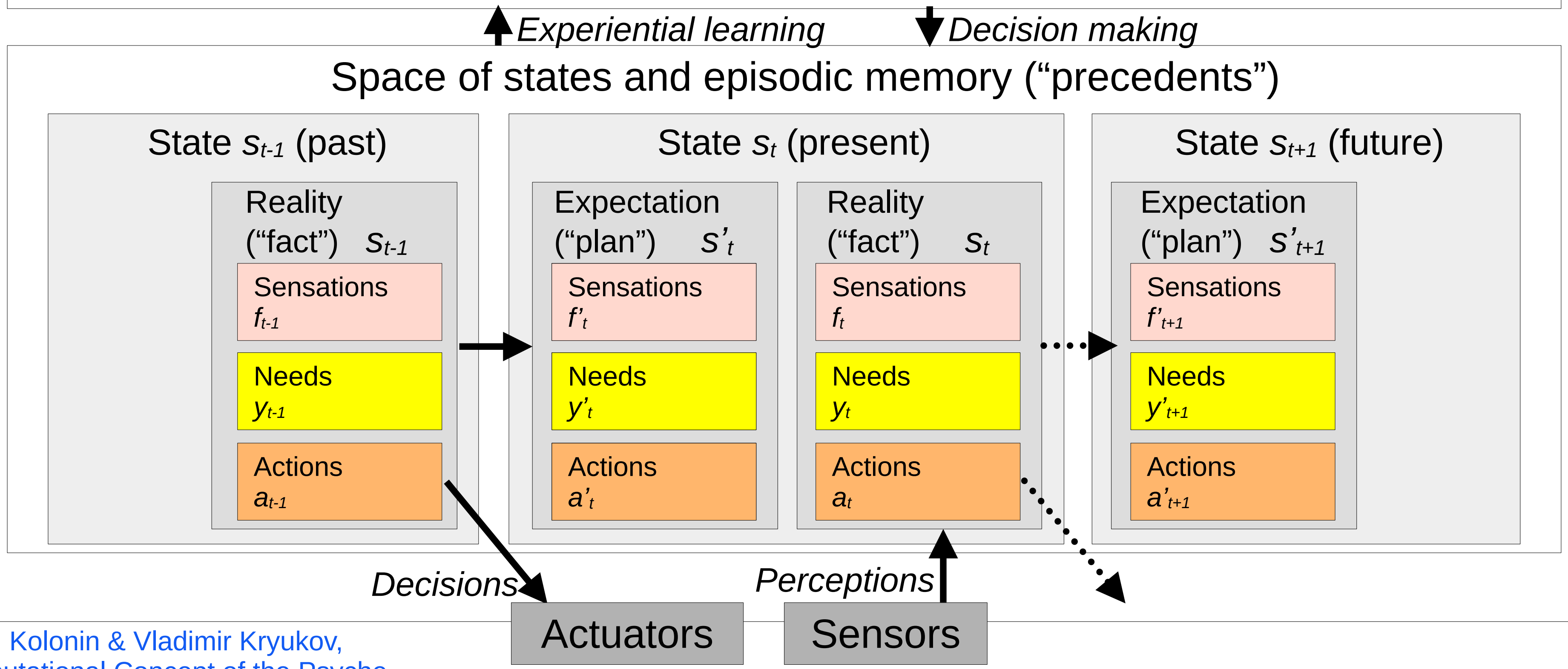
Psyche = Operating system

Intelligence = Decision making system

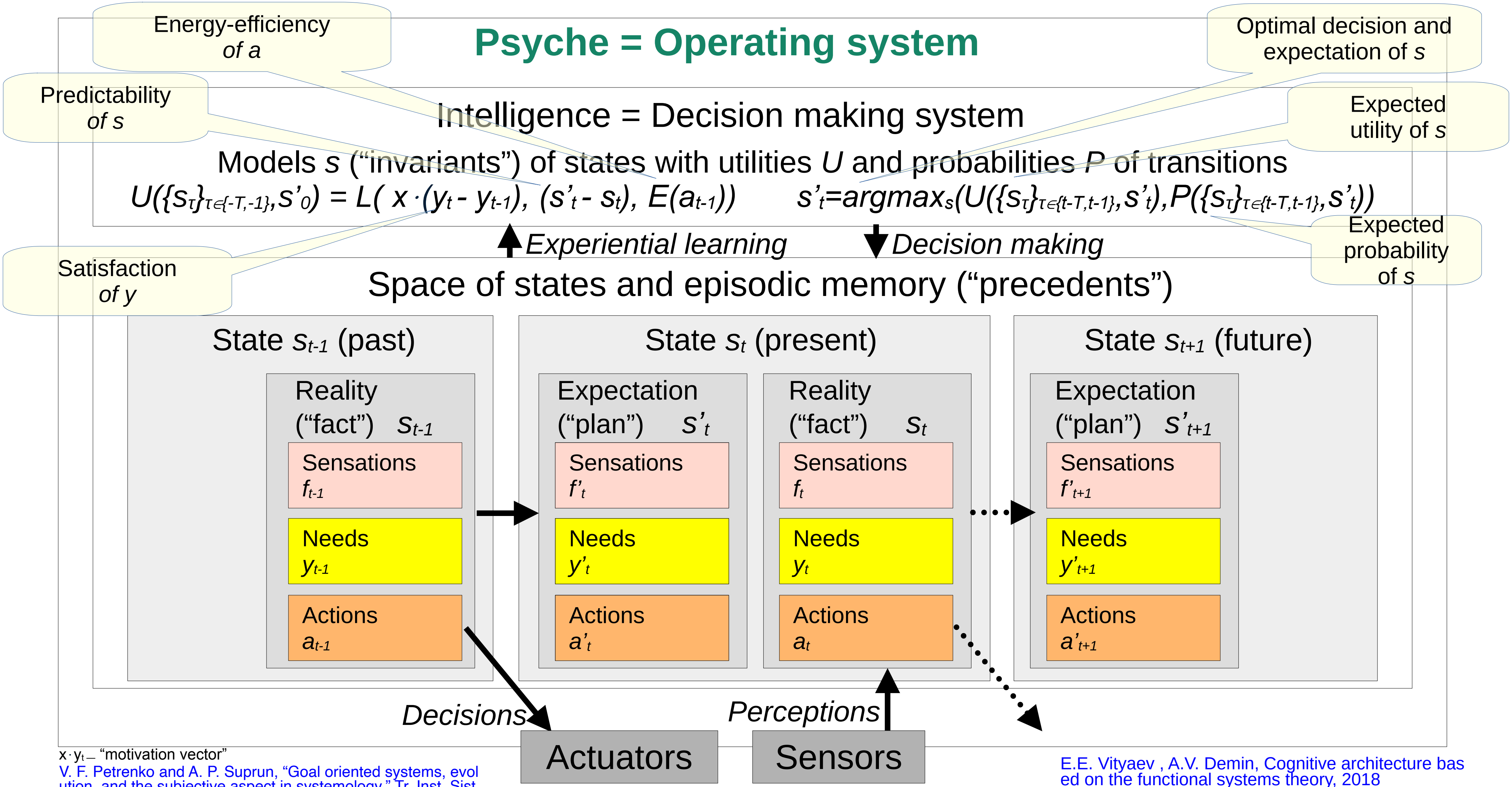
Models s (“invariants”) of states with utilities U and probabilities P of transitions

$$U(\{s_\tau\}_{\tau\in\{-T,-1\}}, s'_0) = L(\, x \cdot (y_t - y_{t-1}),\ (s'_t - s_t),\ E(a_{t-1}))$$

$$s'_t = \operatorname{argmax}_s(U(\{s_\tau\}_{\tau\in\{t-T,t-1\}}, s'_t), P(\{s_\tau\}_{\tau\in\{t-T,t-1\}}, s'_t))$$



Anton Kolonin & Vladimir Kryukov, Computational Concept of the Psyche, https://arxiv.org/abs/2603.15586



$x\cdot y_t$ – “motivation vector”

V. F. Petrenko and A. P. Suprun, “Goal oriented systems, evolution, and the subjective aspect in systemology,” Tr. Inst. Sistem. Analiza RAN 62 (1) (2012)

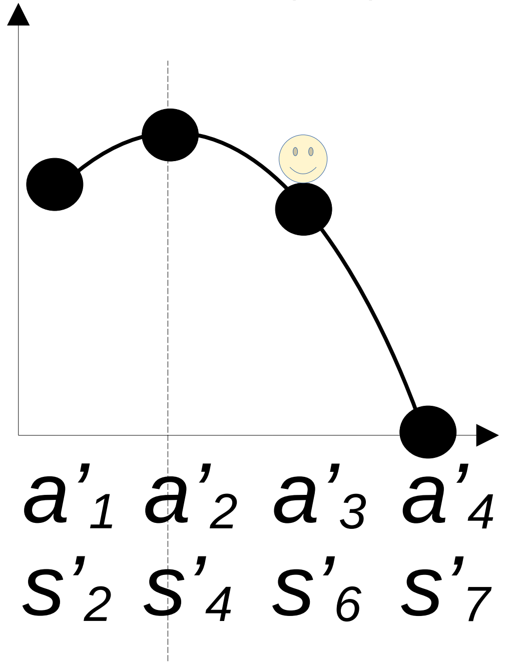
E.E. Vityaev , A.V. Demin, Cognitive architecture based on the functional systems theory, 2018 https://doi.org/10.1016/j.procs.2018.11.072

Decision making as operational risk management

		s'_{t+1}					
s_t	s'_{t+1}	a'	y'	f'	U	P	$\sum U*P$
s_1	s'_2	a'_1	y'_1	…	1.0	0.5	**0.7**
s_1	s'_3	a'_1	y'_2	…	0.4	0.5	
s_1	s'_4	a'_2	y'_3	…	1.0	0.8	**0.8**
s_1	s'_5	a'_2	y'_4	…	0.0	0.2	
s_1	s'_6	a'_3	y'_5	…	0.6	1.0	**0.6**
s_1	s'_7	a'_4	y'_6	…	0.0	1.0	**0.0**

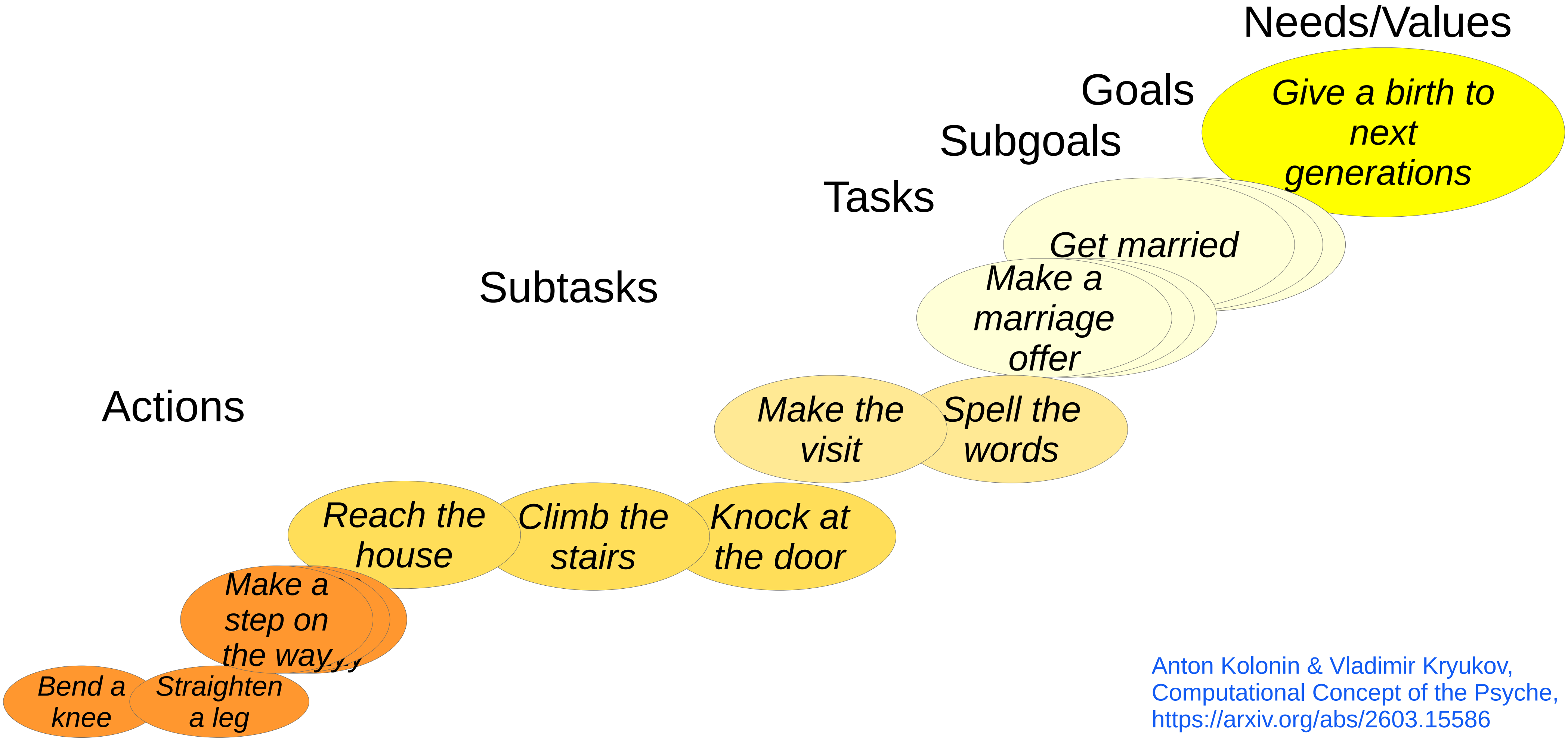
$s'_t = argmax_s(U,P)$

$a'_t = argmax_a(U,P)$



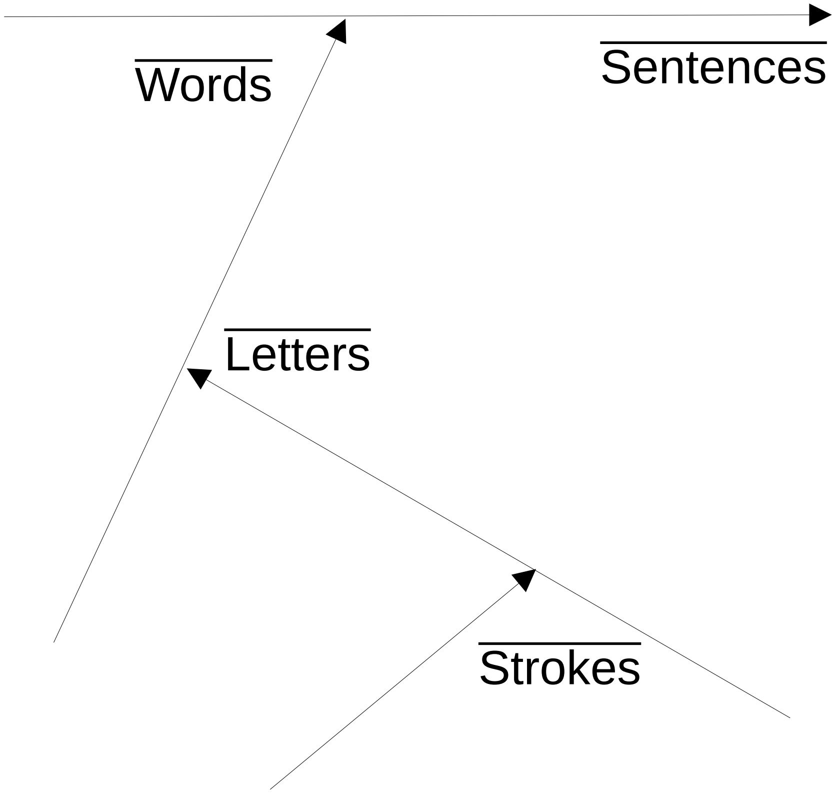
Tversky & Kahneman: most people choose a'_3 и s'_6 (*"smaller profit with greater reliability"*)

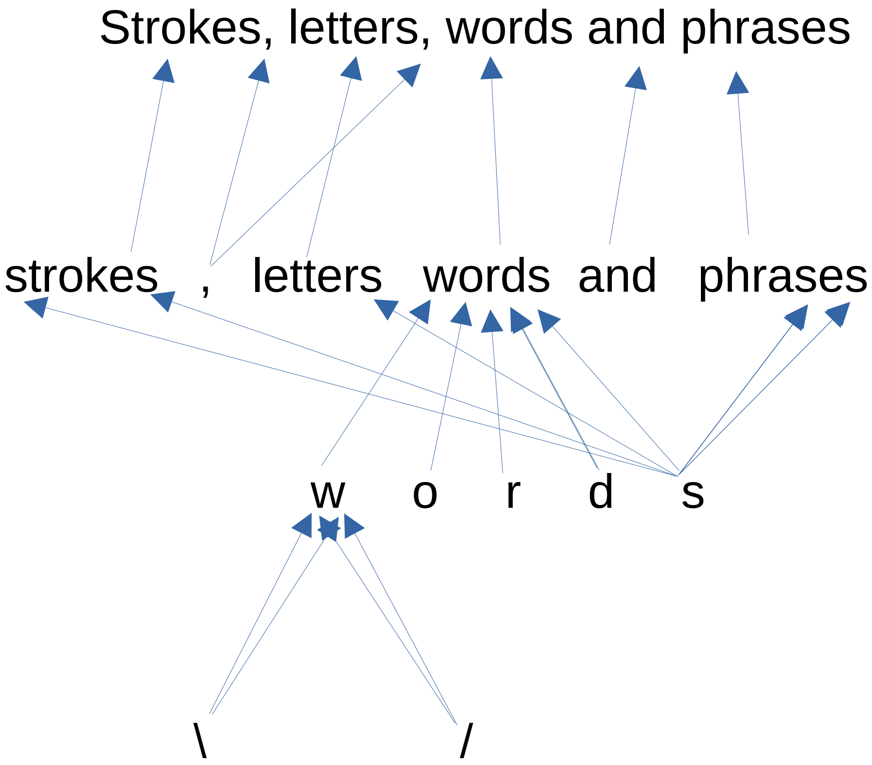
Heterarchy of values/goals/subgoals/tasks/subtasks



Anton Kolonin & Vladimir Kryukov, Computational Concept of the Psyche, https://arxiv.org/abs/2603.15586

Functional equivalence of ~~neural network~~ tensor and (hyper)graph models

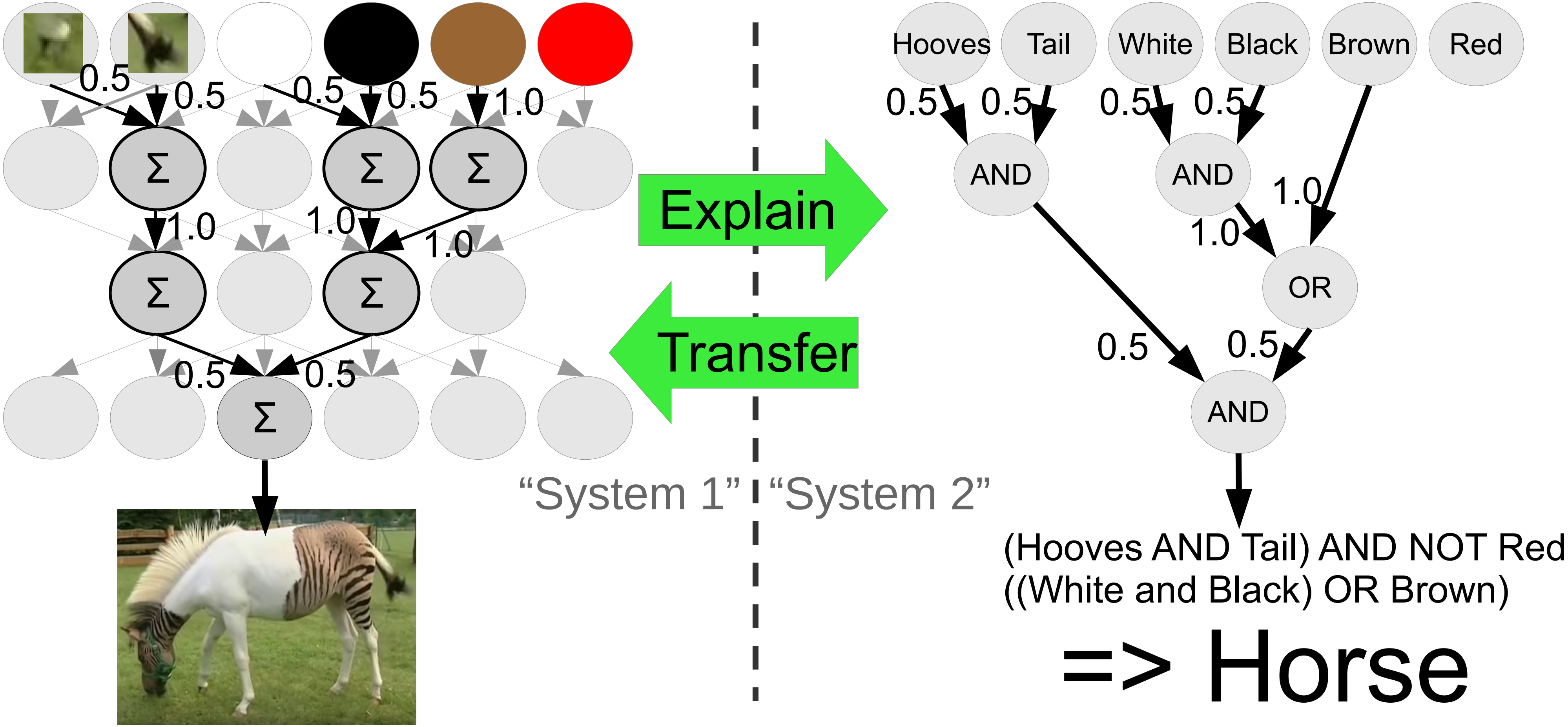




Graph vertex - dimension
Hyper-graph edge - vector
Hyper-graph – tensor space

Pedro Domingos, Tensor Logic: The Language of AI
https://arxiv.org/pdf/2510.12269

Neuro-Symbolic Integration for Interpretable AI



https://www.springerprofessional.de/neuro-symbolic-architecture-for-experiential-learning-in-discret/20008336

Typed tensor logic for different kinds of AI-s (logical, sub-symbolic, probabilistic/non-axiomatic)

Truth-Value Tensor
(NARS/PLN/.../ours)

Property **0.0123456 =750/60750**

striped

horse

Subject

Life-long learning?

Numerical Tensor
(ANN/Bayesian Logic)

Property **~0.01**

striped

horse

Subject

Boolean Tensor
(Boolean Logic)

Property **False**

striped

horse

Subject

Pei Wang: Non-Axiomatic Logic
https://www.worldscientific.com/worldscibooks/10.1142/14486

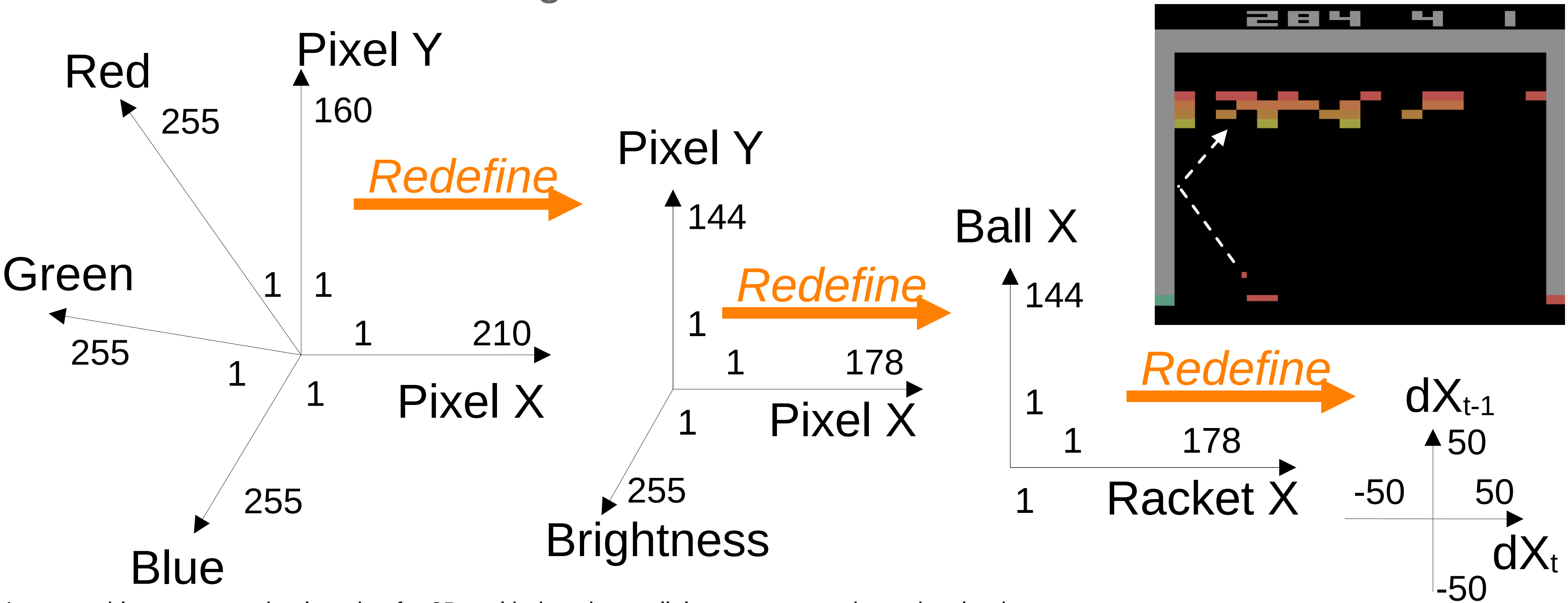
Pedro Domingos, Tensor Logic: The Language of AI
https://arxiv.org/pdf/2510.12269

Problem of dimensionality (reduction) and discreteness (increase)

"What You Need is What you See"

Problem of dimensionality (reduction) and discreteness (increase)

Re-defining environment in Atari Breakout

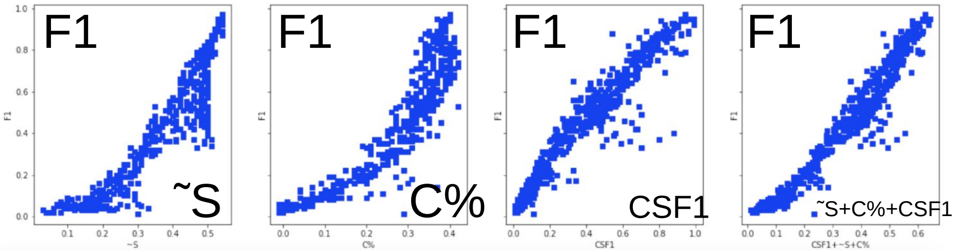


Interpretable representation learning for 3D multi-piece intracellular structures using point clouds

https://www.nature.com/articles/s41592-025-02729-9

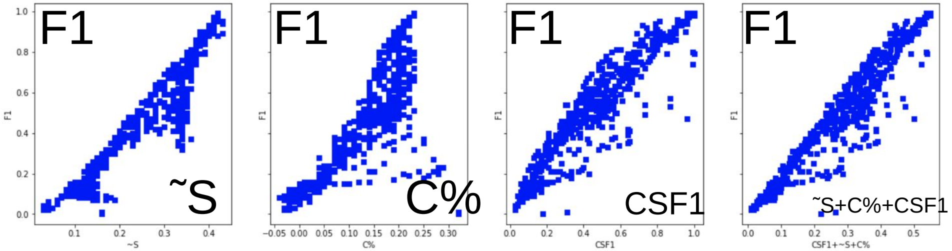
Unsupervised tokenization study: language lexicon as a compression dictionary

English, Train: Brown, Test: Brown 1000

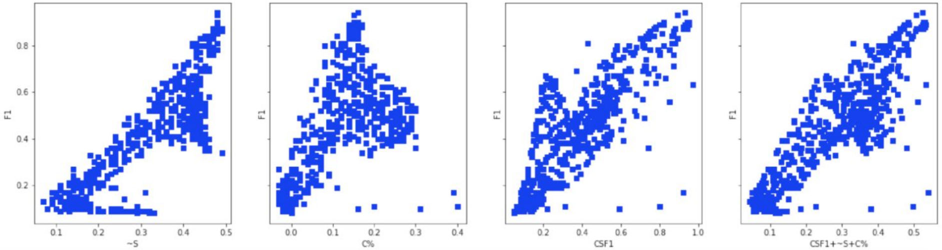


https://github.com/aigents/pygents/blob/main/notebooks/nlp/english/en_tokenizer_auto.ipynb

English, Train: Brown, Test: MagicData 100

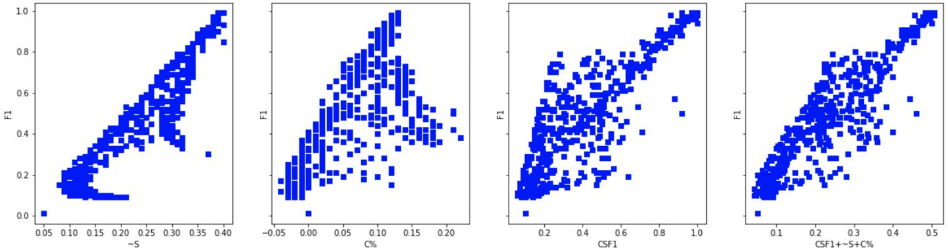


Russian, Train: RusAge, Test: RusAge 1000

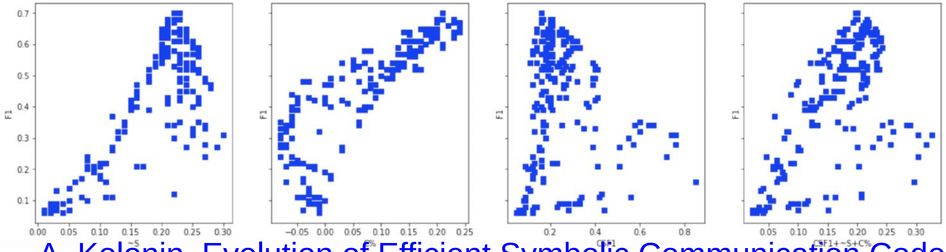


https://github.com/aigents/pygents/blob/main/notebooks/nlp/russian/ru_tokenizer_auto.ipynb

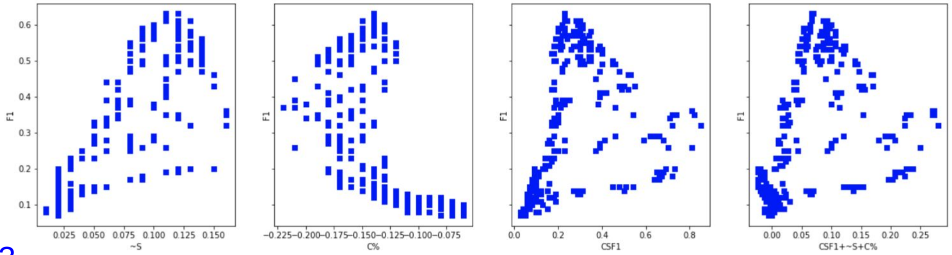
Russian, Train: Brown, Test: MagicData 100



Chinese, Train: CLUE News, Test: CLUE News 1000



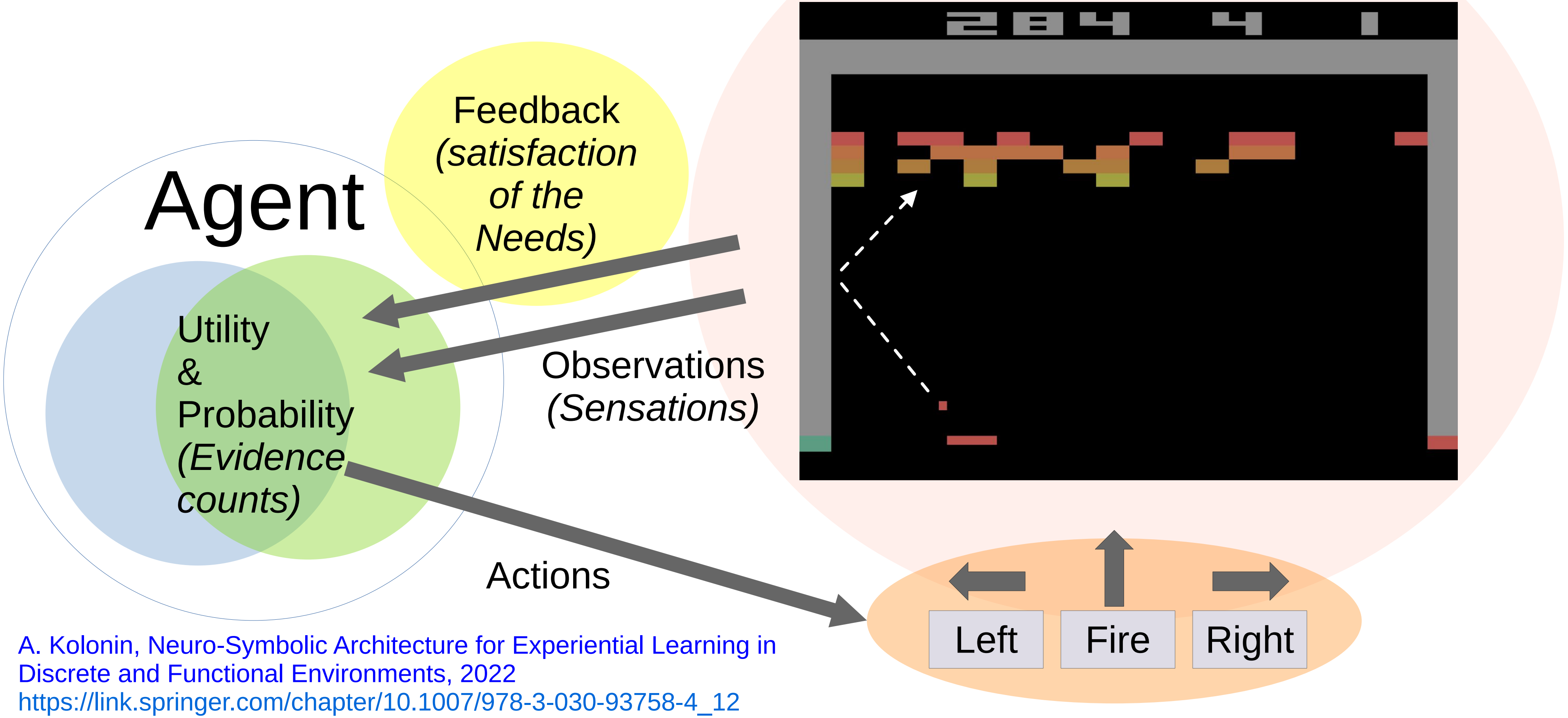
Chinese, Train: Brown, Test: MagicData 100



A. Kolonin, Evolution of Efficient Symbolic Communication Codes, 2023
https://link.springer.com/chapter/10.1007/978-3-031-44865-2_1

Experiential learning study

Virtual gaming environment: OpenAI Gym (Atari Breakout)

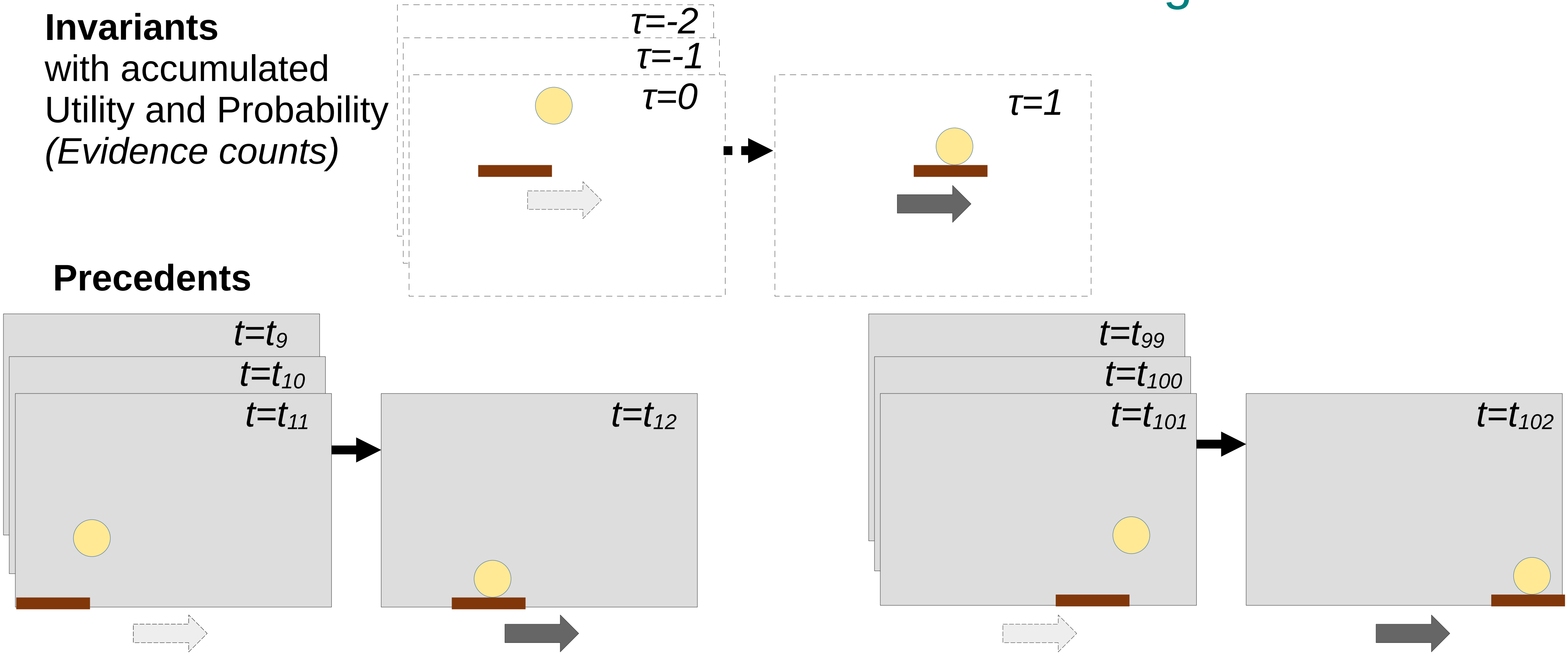


A. Kolonin, Neuro-Symbolic Architecture for Experiential Learning in Discrete and Functional Environments, 2022
https://link.springer.com/chapter/10.1007/978-3-030-93758-4_12

Experiential learning and decision making
based and transitions between state series and “global feedback”

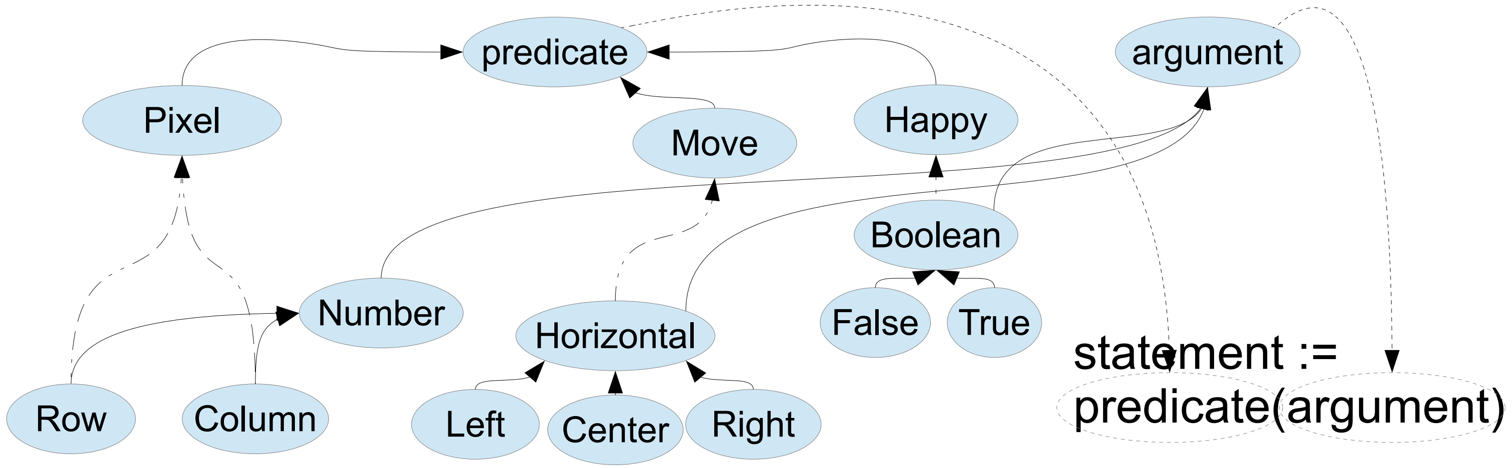
Invariants
with accumulated
Utility and Probability
(Evidence counts)

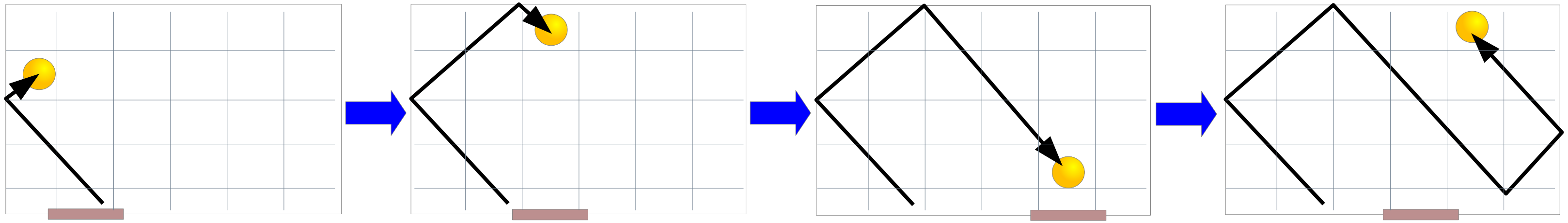
Precedents



A. Kolonin, Neuro-Symbolic Architecture for Experiential Learning in Discrete and Functional Environments, 2022
https://link.springer.com/chapter/10.1007/978-3-030-93758-4_12

Experiential learning - playing “Pong” (Pixels)





Pixel(1,0)
Pixel(4,1)
Happy(False)
=>
Move(Left)

Pixel(0,2)
Pixel(4,2)
Happy(False)
=>
Move(Right)

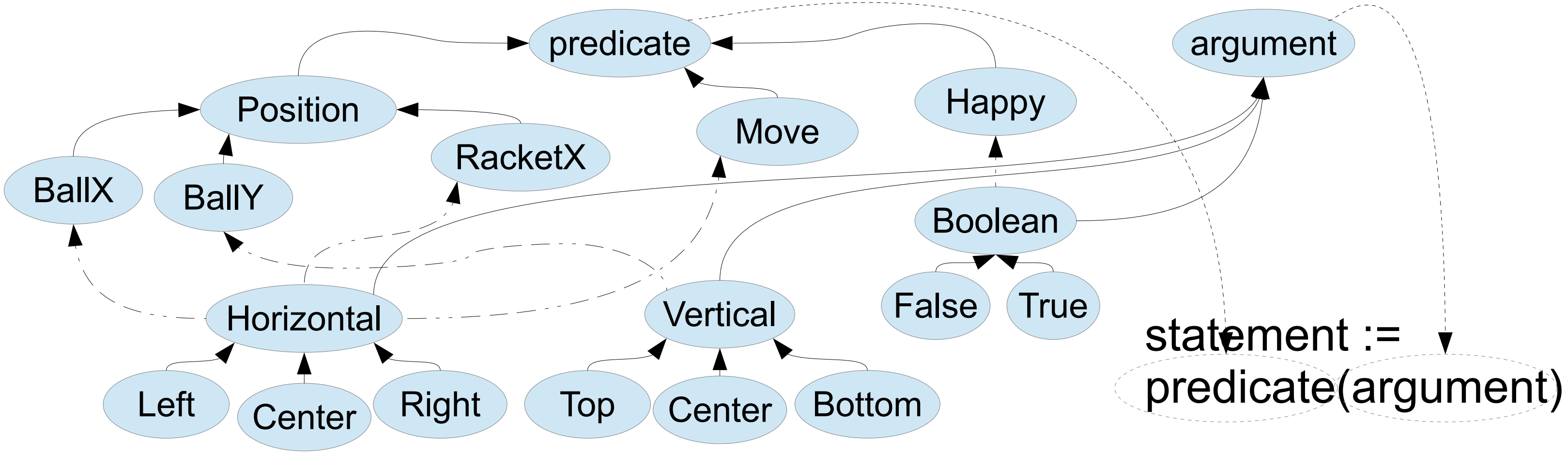
Pixel(3,4)
Pixel(4,4)
Happy(False)
=>
Move(Left)

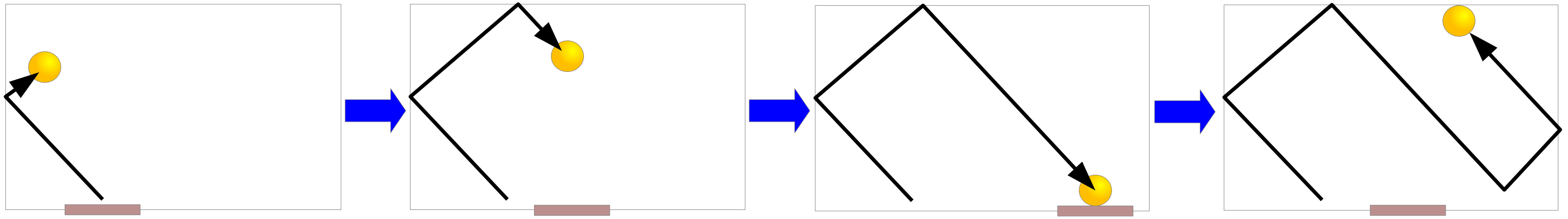
Pixel(0,4)
Pixel(4,3)
Happy(True)
=>
Move(Left)

Global
Feedback

https://www.youtube.com/watch?v=2LPLhJKh95g

Experiential learning - playing “Pong” (Objects)





BallY(Top)
BallX(Left)
RacketX(Left)
Happy(False)
=> **Move(Left)**

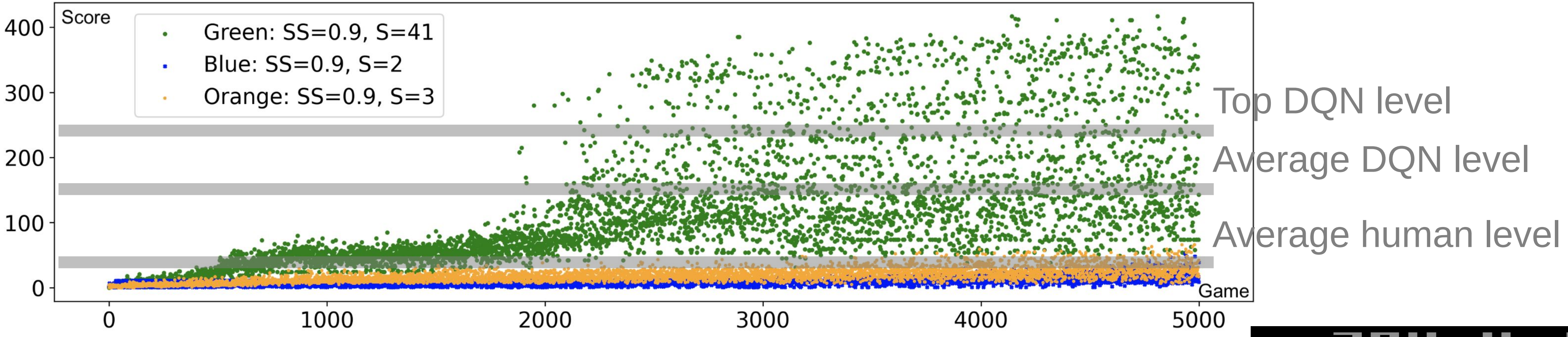
BallY(Top)
BallX(Center)
RacketX(Center)
Happy(False)
=> **Move(Right)**

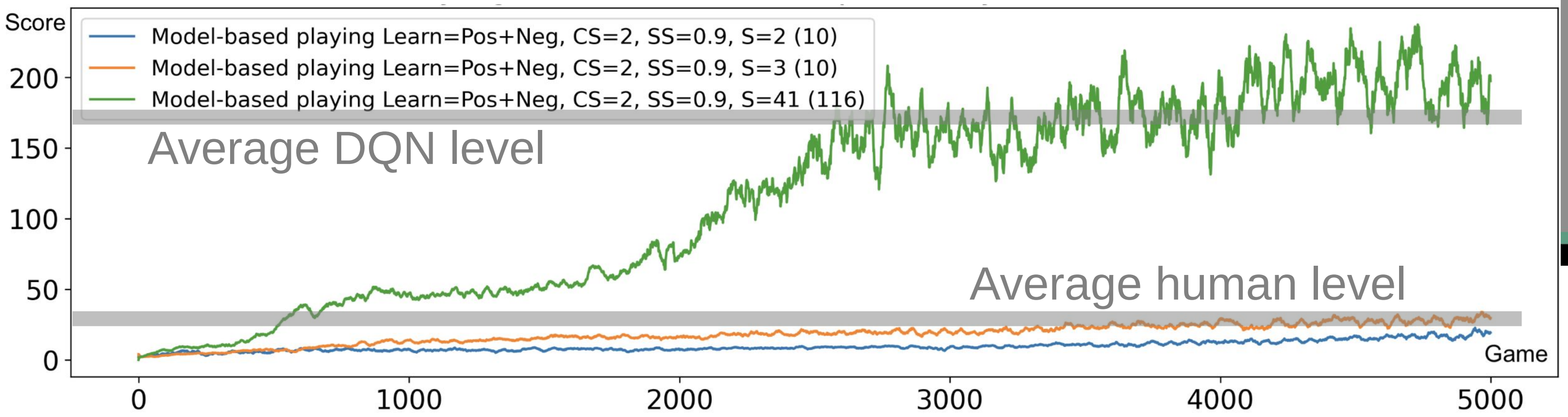
BallY(Bottom)
BallX(Right)
RacketX(Right)
Happy(False)
=> **Move(Right)**

BallY(Bottom)
BallX(Right)
RacketX(Right)
Happy(True)
=> **Move(Left)**

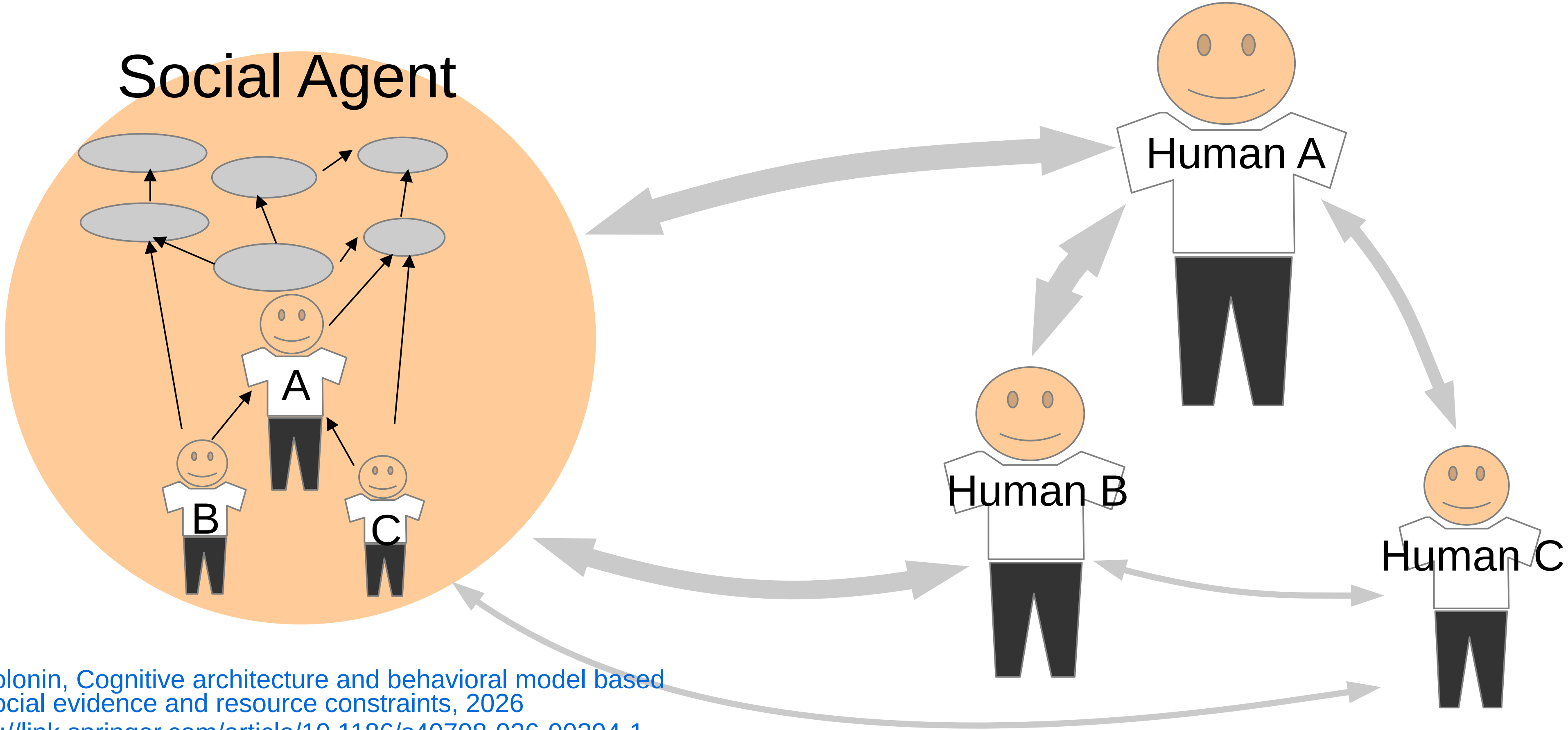
Global
Feedback

Reinforcement learning – experiential learning and decision making based and transitions between state series and “global feedback”



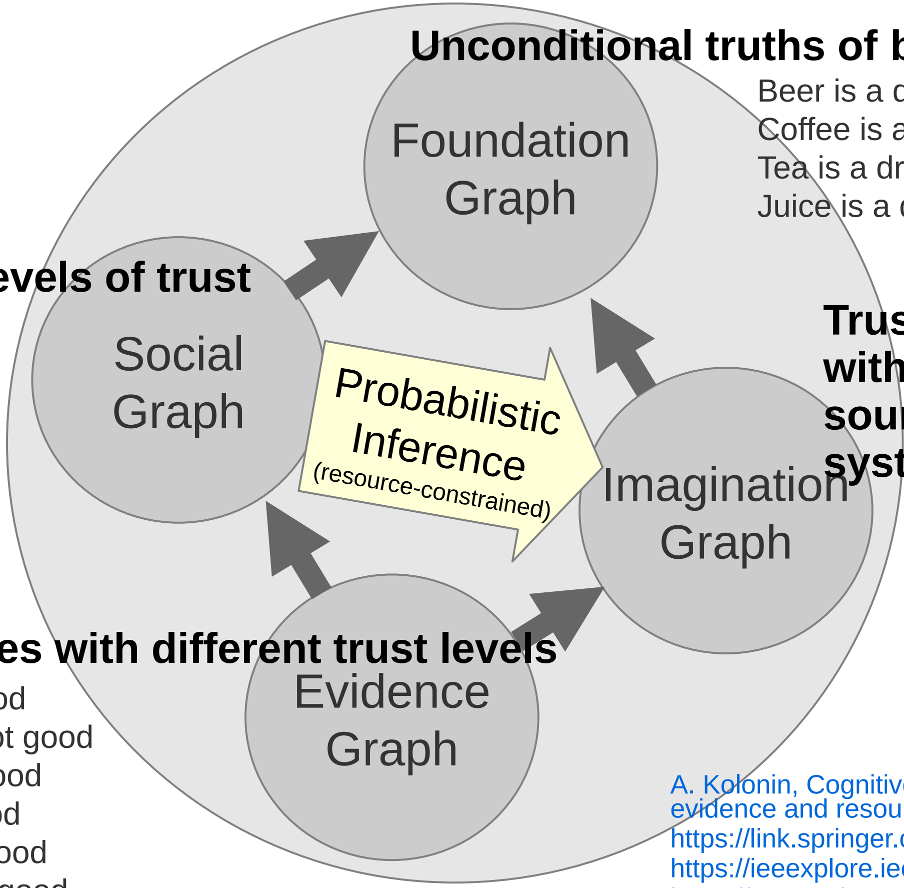


Cognitive architecture and behavioral model based on social evidence and resource constraints



A. Kolonin, Cognitive architecture and behavioral model based on social evidence and resource constraints, 2026
https://link.springer.com/article/10.1186/s40708-026-00294-1

Cognitive architecture and behavioral model based on social evidence and resource constraints



Unconditional truths of belief system

Beer is a drink
Coffee is a drink
Tea is a drink
Juice is a drink

Social connections with levels of trust

Mary is my best friend
Peter is my friend
Bob is Peter's friend

Trusted facts and planned actions, with account to trust levels for sources and justification by belief system

★★★ I like juice very much, will order
★★ I like tea, may order
★ I may try coffee
I won’t try beer

Facts from different sources with different trust levels

Bob said beer is good
Mary said beer is not good
Mary said juice is good
Mary said tea is good
Peter said juice is good
Peter said coffee is good

A. Kolonin, Cognitive architecture and behavioral model based on social evidence and resource constraints, 2026
https://link.springer.com/article/10.1186/s40708-026-00294-1
https://ieeexplore.ieee.org/document/7361869
https://www.sciencedirect.com/science/article/pii/S1877050916317239
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

2015-2026

Cognitive architecture and behavioral model based on social evidence and resource constraints

Unconditional truths of belief system

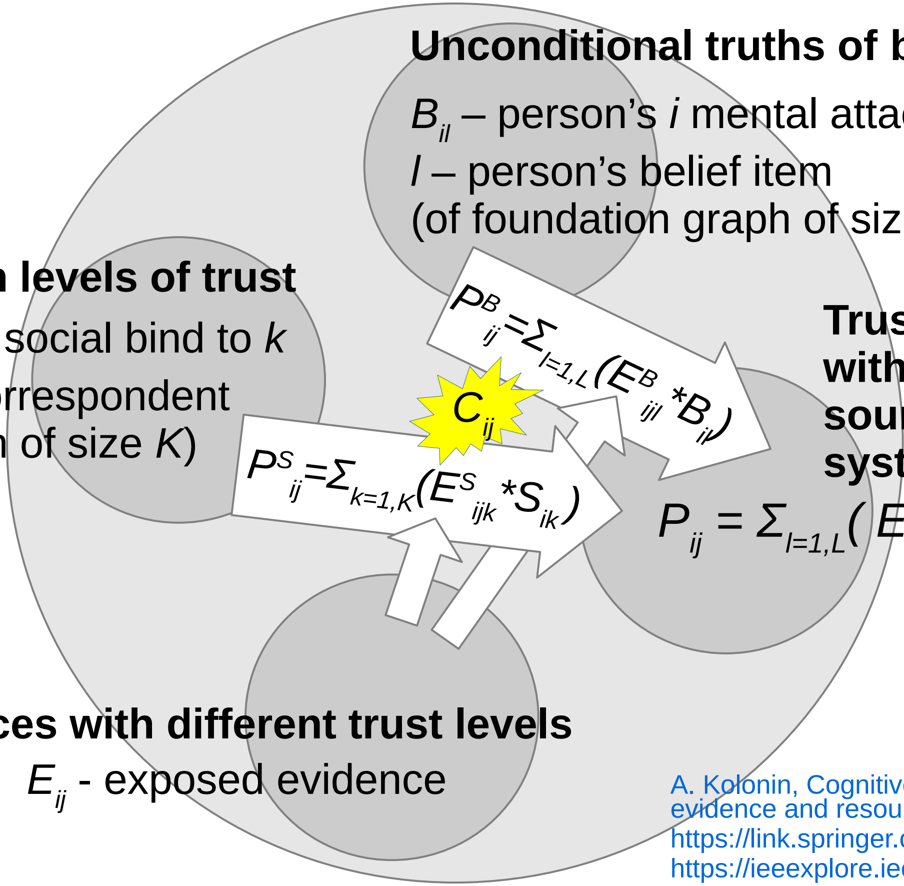
B_{il} – person's *i* mental attachment to *l*

l – person's belief item
(of foundation graph of size *L*)

Social connections with levels of trust

S_{ik} – person's *i* social bind to *k*

k – person's correspondent
(of social graph of size *K*)



Trusted facts and planned actions, with account to trust levels for sources and justification by belief system

$$P_{ij} = \Sigma_{l=1,L}(E^B_{ijl} * B_{il}) * \Sigma_{k=1,K}(E^S_{ijk} * S_{ik})$$

Facts from different sources with different trust levels

E_{ij} - exposed evidence

A. Kolonin, Cognitive architecture and behavioral model based on social evidence and resource constraints, 2026
https://link.springer.com/article/10.1186/s40708-026-00294-1
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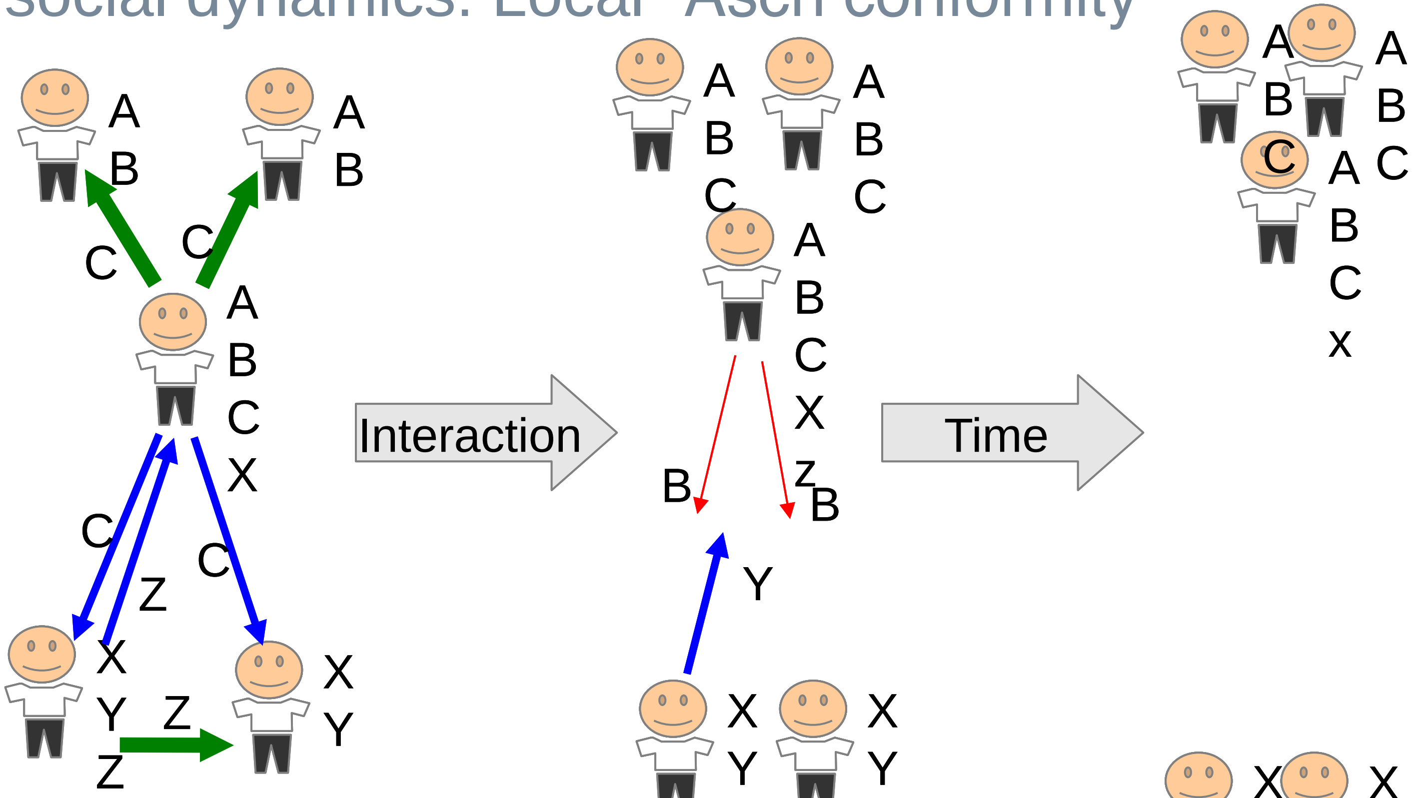
2015-2026

Cognitive architecture and behavioral model based on social evidence and resource constraints

Predicted social dynamics: Local ”Asch conformity”



https://link.springer.com/article/10.1186/s40708-026-00294-1
https://ieeexplore.ieee.org/document/7361869
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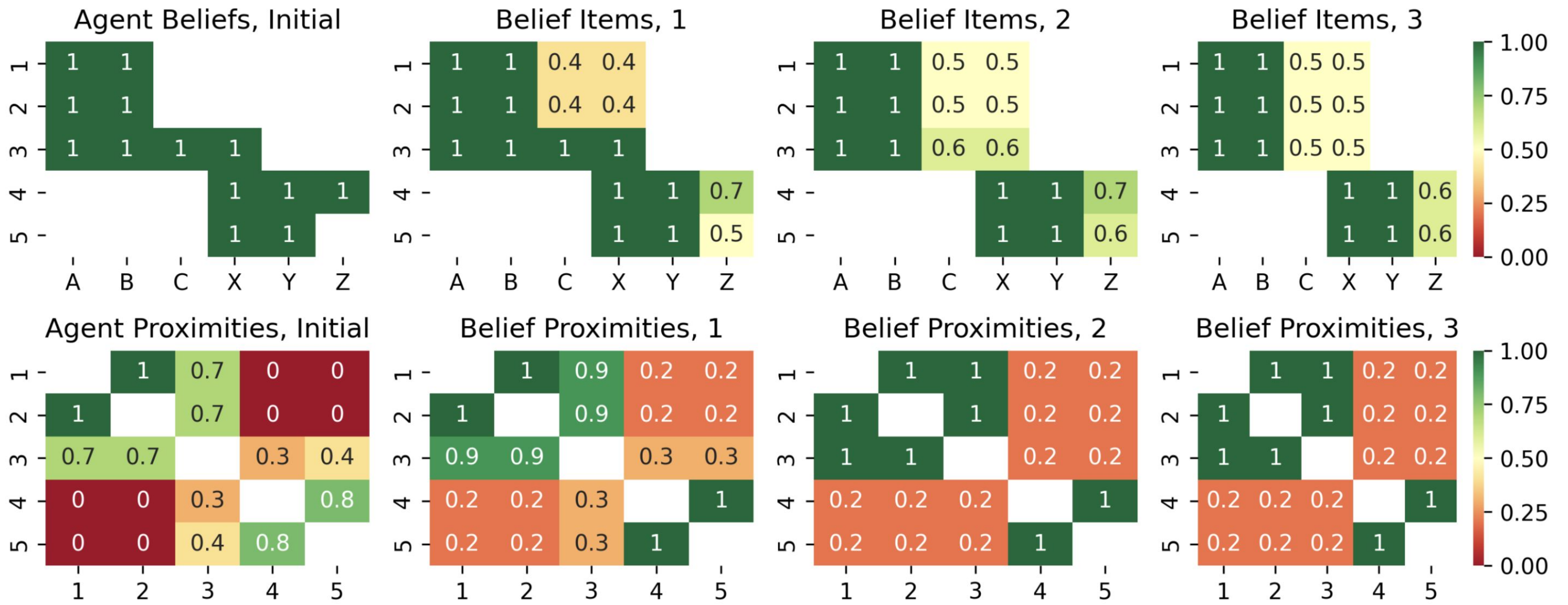
https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

Simulation: socially limited, cognitively capable community



Results of three rounds of multi-agent simulation with the forgetting threshold peer threshold = 0.5, the social relationship threshold forgetting threshold = 0.0 (limiting social connections K without constraint on belief capacity L). Top row: values of belief items A, B, C, X, Y, Z for five agents 1, 2, 3, 4, 5. Bottom row: social proximity matrices between agents, estimated as the cosine similarity of their beliefs. From left to right: the initial state before the simulation, and then the updated states of beliefs and proximities after three subsequent rounds of multi-agent communication (”**Asch conformity**”).

Thank you for attention! Questions?

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Full version of the presentation video

Our latest paper